

3. Visualize the data using Python by plotting the graphs for assignment no. 1 and 2. Consider a suitable data set. Use Scatter plot, Bar plot, Box plot, Pie chart, Line Chart.

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```
import pandas as pd
import matplotlib.pyplot as plt
import kagglehub
import os

data = {
    "Age": [18, 19, 20, 21, 22, 23, 24, 25, 26, 27],
    "Study_Hours": [2, 3, 4, 5, 6, 4, 3, 7, 6, 8],
    "Marks": [45, 50, 55, 65, 70, 60, 58, 75, 72, 80],
    "Attendance": [60, 65, 70, 75, 80, 72, 68, 85, 82, 90],
    "Result": [0, 0, 0, 1, 1, 1, 0, 1, 1, 1] # 0 = Fail, 1 = Pass
}

df = pd.DataFrame(data)
df
```

```
plt.figure()
plt.scatter(df["Age"], df["Marks"])
plt.xlabel("Age")
plt.ylabel("marks")
plt.title("Relationship between age and attend")
plt.show()
```

Start coding or [generate](#) with AI.

```
path = kagglehub.dataset_download("johnsmith88/heart-disease-dataset")
```

```
file_path = os.path.join(path, "heart.csv")  
df = pd.read_csv(file_path)
```

```
df.head()
```

```
#Relationship between age abd cholestrol  
plt.figure()  
plt.scatter(df["age"], df["chol"])  
plt.xlabel("Age")  
plt.ylabel("Cholesterol")  
plt.title("Scatter Plot: Age vs Cholesterol")  
plt.show()
```

```
#Average cholestrol for heart disease vs/ no disease  
avg_chol = df.groupby("target")["chol"].mean()  
  
plt.figure()  
avg_chol.plot(kind="bar")  
plt.xlabel("Heart Disease (0 = No, 1 = Yes)")  
plt.ylabel("Average Cholesterol")  
plt.title("Bar Plot: Average Cholesterol vs Heart Disease")  
plt.show()
```

Patients with heart disease show different average cholesterol levels.

```
#Distribution of age bp cholestrol
plt.figure()
df[["age", "trestbps", "chol"]].boxplot()
plt.ylabel("Values")
plt.title("Box Plot: Distribution of Heart Disease Parameters")
plt.show()
```

Displays median, spread, and outliers of numerical attributes.

```
#Heart Disease Distribution
disease_count = df["target"].value_counts()

plt.figure()
plt.pie(disease_count, labels=["No Disease", "Disease"], autopct="%1.1f%%")
plt.title("Pie Chart: Heart Disease Distribution")
plt.show()
```

Shows proportion of patients with and without heart disease.

```
# Sort by age for smooth lines
df_sorted = df.sort_values("age")

# Create Figure and Axes (OO style)
fig, ax = plt.subplots(figsize=(6, 3), layout='constrained')

# Multiple line plots with different colors & styles
ax.plot(df_sorted["age"], df_sorted["chol"],
        color='blue', linestyle='--', label='Cholesterol')

ax.plot(df_sorted["age"], df_sorted["trestbps"],
        color='green', linestyle='--', label='Resting BP')

ax.plot(df_sorted["age"], df_sorted["thalach"],
        color='red', linestyle=':', label='Max Heart Rate')

# Labels & title
ax.set_xlabel("Age")
ax.set_ylabel("Values")
ax.set_title("Multiple Line Plot: Heart Parameters vs Age")

# Legend
ax.legend()

plt.show()
```

Multiple parameters vary differently with age.

Max heart rate decreases as age increases.

Double-click (or enter) to edit